Jatayu's Last Stand

Category: Forensics | **Flag:** `PINAKA{jatayu_saw_what_th3_r3plays_c0uld_n0t_RAVANA_T0OK_SITA}`

Overview

Three artifacts: a telemetry ROM, a flash image and a radio I/Q capture. The ROM holds a TLV calibration block hidden behind an LFSR. The flash pages are protected with CRC-16/CCITT-FALSE. The committed head of the flash journal tells you which warning in the radio traffic is the real one (`CHC95`).

`solve.py` decrypts the calibration, checks every flash page and derives the committed head. The 16-word warning from the radio is a fixed input to the script; the BPSK demodulation isn't part of `solve.py`.

What came out

- calibration scale: `0.0004894041921943426`
- calibration offset: `-489.404296875`
- flash pages: 9 valid, 0 invalid
- committed `CHC95` head: `0654b64ed636840b`
- warning: `GTUZQ 3P948 NPGZK U9NTA XG7W8 HDGAS HJEYA CHC95 M49J3 4F4VG CZ5H6 C8Z3R TXY77 MJ98B 6LQD2 FAXJ9`

```
python3 solve.py
```

Result

The validator answered:


```
{"flag":"PINAKA{jatayu_saw_what_th3_r3plays_c0uld_n0t_RAVANA_T0OK_SITA}","ok":true}
```


The validator was hosted on Render and has since been suspended by its owner, so this can't be re-submitted right now. If it comes back:

```
python3 solve.py --submit
```

```
/home/darknotch/CTF/Pinaka/FOR_Jatayu_player
Permissions Size User      Date Modified Name
.rw-rw-r--  6.1k darknotch 27 Oct 22:07  02-Jatayu/acquisition/flash.img
.rw-rw-r--  104k darknotch 27 Oct 22:07  02-Jatayu/acquisition/radio.iq
.rw-rw-r--    55 darknotch 27 Oct 22:07  02-Jatayu/acquisition/telemetry.rom

…/Pinaka/FOR_Jatayu_player ❯
```

```
Telemetry calibration: {"scale": 0.0004894041921943426, "offset": -489.404296875, "tlv_tags": ["0x01", "0x02", "0x10", "0x11", "0x12", "0x13
 "flash_pages_valid": 9, "flash_pages_bad": 0, "CHC95_page": 9}
Recovered radio words: GTUZQ 3P948 NPGZK U9NTA XG7W8 HDGAS HJEYA CHC95 M49J3 4F4VG CZ5H6 C8Z3R TXY77 MJ98B 6LQD2 FAXJ9
Flash commit head: 0654b64ed636840b
Radio capture bytes: 104248

…/Pinaka/FOR_Jatayu_player ❯
```


```
{
  "words": [
    "GTUZQ",
    "3P948",
    "NPGZK",
    "U9NTA",
    "XG7W8",
    "HDGAS",
    "HJEYA",
    "CHC95",
    "M49J3",
    "4F4VG",
    "CZ5H6",
    "C8Z3R",
    "TXY77",
    "MJ98B",
    "6LQD2",
    "FAXJ9"
  ],
  "warning": "GTUZQ 3P948 NPGZK U9NTA XG7W8 HDGAS HJEYA CHC95 M49J3 4F4VG CZ5H6 C8Z3R TXY77 MJ98B 6LQD2 FAXJ9",
  "commit_head": "0654b64ed636840b",
  "scale": 0.0004894041921943426,
  "offset": -489.404296875
}
```


```
…/CTF/Writeups ❯ curl -sS -X POST https://pinaka-jatayu-validator.onrender.com/submit \
 -H 'Content-Type: application/json' -d '{
 "words":["GTUZQ","3P948","NPGZK","U9NTA","XG7W8","HDGAS","HJEYA","CHC95",
          "M49J3","4F4VG","CZ5H6","C8Z3R","TXY77","MJ98B","6LQD2","FAXJ9"],
 "warning":"GTUZQ 3P948 NPGZK U9NTA XG7W8 HDGAS HJEYA CHC95 M49J3 4F4VG CZ5H6 C8Z3R TXY77 MJ98B 6LQD2 FAXJ9",
 "commit_head":"0654b64ed636840b",
 "scale":0.0004894041921943426,
 "offset":-489.404296875}'
{"flag":"PINAKA{jatayu_saw_what_th3_r3plays_c0uld_n0t_RAVANA_T00K_SITA}","ok":true}

…/CTF/Writeups ❯
```

```
Telemetry calibration: {"scale": 0.0004894041921943426, "offset": -489.404296875, "tlv_tags": ["0x01", "0x02", "0x10", "0x11", "0x12", "0x13", "0x14", "0x15", "0xff"],
 "flash_pages_valid": 9, "flash_pages_bad": 0, "CHC95_page": 9}
Recovered radio words: GTUZQ 3P948 NPGZK U9NTA XG7W8 HDGAS HJEYA CHC95 M49J3 4F4VG CZ5H6 C8Z3R TXY77 MJ98B 6LQD2 FAXJ9
Flash commit head: 0654b64ed636840b
Radio capture bytes: 104248

…/Pinaka/FOR_Jatayu_player ❯
```